

FORDEC · Engine Fire after V1, Fire Continues

SXS2UM (XQ945) · B737-800 · EHAM RWY 18L → LTBJ · OPC-6 · SunExpress
FORDEC model (OF/Standardization, rev. 00)

Point of decision: EOSID AMSX4, 3000 ft, clean, MCT
Engine Fire checklist and After Takeoff checklist complete
Commander retains full authority

F FACTS · What's the matter?

TIME CRITICAL: YES Inextinguishable engine fire. Company model: gather options that **minimise in-flight time** with an acceptable degree of safety, **choose the first acceptable option**, focus on execution and land as soon as possible.

AIRCRAFT

- ENG 2 fire warning at 137 kt (V1 132). Takeoff continued (OM-B 3.10).
- Memory items done. Bottle 1 and bottle 2 discharged. **Fire switch 2 still illuminated.**
- ENG 2 shut down. ENG 1 at MCT, set by hand: **A/T INOP (MEL 22-04).**
- Weight ≈ **69.3 t** vs MLW **66,360 kg** → overweight ≈ 3 t. Fuel is not a limit.
- OEI: **CAT I only**. Autoland needs both engines (OM-B 1.5.1). OEI landing **flaps 15** (no EO flaps 30 planned, OM-B 1.9.3.2).
- Engine anti-ice ON, OAT −4 °C: **cold temperature corrections** on own-navigation altitudes.

EHAM WEATHER AND RUNWAYS

- METAR EHAM 151325Z 12022KT 1000 R18C/0250N R18L/0200N R06/1000N SN BLSN BKN005 OVC020 M04/M05 Q1025 BECMG 0700 +SN BLSN

 - Trend **deteriorating**: 0700 +SN BLSN. TAF keeps 0700 +SN until about 17Z.
 - RWY 06 RWYCC 4 (arrival runway) · 18L RWYCC 4 · 18C RWYCC 2 · **all other runways closed** (snow).
 - Company crosswind limits (OM-B 1.2.4.6.1): RWYCC 4 **20 kt** incl. gust · RWYCC 2 **5 kt**. Minus **5 kt** when asymmetric reverse is used on a contaminated runway (note 7).

WIND COMPONENTS, 120/22

Runway (QFU)	Angle	Crosswind	Head / tail	Limit	Result
06 (055°)	65°	$22 \times \sin 65^\circ \approx 19.9$ kt	+9 kt head	20 kt · 15 kt with asym. reverse	At the limit → no reverse
18L (181°)	61°	≈ 19.2 kt	+11 kt head	20 kt	RVR 200 m below CAT I
18C (181°)	61°	≈ 19.2 kt	+11 kt head	5 kt (RWYCC 2)	Crosswind 4× the limit

O OPTIONS · What kind of possibilities do we have?

Option	Distance	Approach / minima	Weather / runway	Status
EHAM RWY 06	≈ 10 min radar vectors	ILS 06 · CAT I (RVR 550 m)	RVR 1000 m · RWYCC 4 · 3,439 m (LDA ≈ 3,190 m) · crosswind ≈ 20 kt	ACCEPTABLE
EHAM RWY 18L	≈ 10 min	ILS · OEI = CAT I only	RVR 200 m	NOT POSSIBLE
EHAM RWY 18C	≈ 10 min	ILS · CAT I	RWYCC 2 · RVR 250 m	NOT POSSIBLE
EHRD Rotterdam 24	24 NM	ILS CAT I	2,200 m · weather unknown, same snow system likely	MARGINAL
EHEH Eindhoven 03/21	56 NM	ILS	3,000 m · weather unknown · RFFS and hours to check	TOO FAR
EBBR Brussels 25L/R	85 NM ≈ 20+ min	OEI = CAT I only	Long runways · weather unknown	TOO FAR
Hold to reduce weight	Burning fuel to MLW takes far too long with a fire that does not go out.			REJECT

Before deciding: ask ATC for the latest RVR 06, wind and runway condition for 06; run the landing performance tool for flaps 15, RWYCC 4, actual weight, **no reverse**, crosswind.

R RISKS / BENEFITS · Which option mitigates the threats best?

Option	Benefits	Risks → mitigation
EHAM 06	Closest. Shortest time in the air with the fire. Runway, ILS and minima meet CAT I. Full rescue and fire service on scene, company handling and maintenance. Crew and ATC know the area.	Crosswind ≈ 20 kt at the RWYCC 4 limit → no reverse (keeps the 20 kt limit), crab/de-crab technique, abort if gusts exceed. Contaminated runway, overweight, flaps 15 (higher Vref) → landing distance from the tool with no reverse, autobrake per result or max manual braking, full length, no hold-off. Visibility falling (+SN) → keep checking RVR ≥ 550 m. Manual thrust (A/T INOP) + OEI → brief speed control and the OEI go-around.
EHRD	Close, CAT I.	Short runway for an overweight flaps 15 landing on a probably contaminated runway. Weather unknown. Smaller rescue cover.
EHEH / EBBR	Longer runways, possibly better weather.	Unknown weather. 15-25 more minutes with an inextinguishable fire: not acceptable in a time-critical case.

D DECISION · What are we going to do?

Return and land EHAM RWY 06 · ILS CAT I · flaps 15 · no reverse · overweight landing

1. First priority, safety: EHAM 06 is the **first acceptable option** and the one with the least time in the air.
2. It is also the most efficient: home of the handling, maintenance and passenger solutions.

Go-around plan: if not stabilised by 1000 ft or no visual reference at DA → OEI go-around, then re-assess (RVR / wind); next option EHEH only if 06 becomes unusable.

E EXECUTE · Who does what, when, and how?

INFORM

ATC (MAYDAY already declared):

SXS2UM, engine fire, fire continues. Request immediate return, vectors ILS 06.
Overweight landing, possible evacuation on the runway.
Request fire services at the runway. POB ____, fuel __ tonnes.

SCCM · TTRR

Type	Engine fire, not extinguished. Emergency landing at Amsterdam, evacuation possible.
Time	About 10 minutes. Synchronise watches.
Risks	Fire on the right side: do not use right exits if fire is seen. Wait for the command. Standby signal / "EVACUATE" or "ATTENTION CREW AT STATIONS" / "CANCEL ALERT".
Re-seating	As required for the exits; passengers away from the right overwing area if possible. Cabin secure in __ min.

PAX PA per SXS-OF-GUI-006. **IOCC** via ACARS or ATC if time allows.

PREPARE

- **FMC:** ILS 06, flaps 15 Vref (+ wind additive within the flap placard), missed approach, EO acceleration.
- **Charts / nav aids:** ILS 06, missed approach, MSA. Identify the ILS.
- **Cold temperature correction** at -4°C on DA and own-navigation altitudes (not under radar vectors).
- **Landing performance tool:** actual weight, flaps 15, RWYCC 4, crosswind, no reverse, autobrake setting.
- **Overweight:** normal profile, no high sink rate, land in the touchdown zone, full length, no early turn-off, **no autoland** (OM-A 8.3.1.19.1, FCTM).

BRIEFING (TEM)

- Threats: fire, crosswind at the limit, contaminated runway, overweight, flaps 15, manual thrust, falling visibility, snow on the windshield.
- Who lands (Commander decides), callouts, stabilised criteria, OEI go-around, **evacuation plan** after stop.

CHECKLISTS

When	What
Now	ONE ENGINE INOPERATIVE LANDING checklist. All deferred items answered by the PF (OM-B 3.10).
Descent / approach	Deferred items: ONE ENGINE INOPERATIVE LANDING DEFERRED DESCENT / APPROACH / LANDING checklists (name them, OM-B 3.12.1).
After stop	Parking brake, assess with fire services → EVACUATION checklist if needed (FO: flaps 40, pressurization MAN, outflow valve OPEN, advise ATC, all fire switches override and pull) or PA "CANCEL ALERT".

Time critical: only essential steps. Skip anything that does not help you land sooner or safer.

C CHECK · Is everything still OK?

Fire	Still burning? Spreading (other warnings, smoke)? If the situation worsens, the plan does not change: land as soon as possible.
Weather at 06	RVR 06 ≥ 550 m, crosswind ≤ 20 kt incl. gust. Ask ATC on every frequency change. Below minima → go-around and re-plan.
Engine 1 / systems	N1, EGT, oil at MCT. Anti-ice ON. Electrical: one generator, APU if started per checklist.
Approach	Stabilised by 1000 ft (landing flaps 15, speed $+10/-5$, 1 dot, ≤ 1000 fpm or the briefed rate). Otherwise go around.
Information	ATC, SCCM and IOCC updated with any change in time or plan. Fire services briefed on the right-engine fire.

The situation can change at any time. If RWY 06 becomes unusable, the next option is the nearest acceptable airfield with known weather. Ask ATC for EHEH / EBBR weather early so the fallback is ready.

Sources: SunExpress FORDEC model (IMG_1058-1059), OM-B 3.10, 1.2.4.6.1 incl. note 7, 1.5.1, 1.5.3.1, 1.9.3.2, 3.12.1; OM-A 8.3.1.16.6, 8.3.1.19.1; OPC-6 data sheet and EHAM AOI. Distances are great-circle from EHAM. Alternate runway lengths and RFFS

categories come from public sources and are not verified against the AIP. Alternate weather is not given in the scenario: ask for it. Weight after takeoff and times are estimates.